

Poojita Kaja

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Education

Florida State University, Tallahassee, FL
M.S. in Interdisciplinary Data Science

Aug 2023 – May 2025
GPA: 3.8 / 4.0

SRM University, Andhra Pradesh, India
B.Tech. in Electronics and Communication Engineering

Jun 2019 – May 2023
GPA: 8.6 / 10

Work Experience

Data Analyst (Government Analyst II)

Jun 2025 – Present

Agency for Health Care Administration (AHCA), Florida

- Cleaned and transformed 10,000+ records across 5 healthcare datasets using R and Python to support data analysis for Florida Medicaid programs, including Graduate Medical Education (GME), TEACH, Startup Bonus, SMRP, and Doctor Slot Allocation.
- Conducted data validation and trend analysis on 12+ program reports, identifying 3 recurring anomalies and helping streamline Medicaid funding review processes.
- Designed and developed 4 interactive dashboards in Tableau to visualize KPIs and allocation trends, improving reporting turnaround time by 25% for internal teams.
- Collaborated with 3 cross-functional teams to automate monthly reporting workflows, ensuring HIPAA compliance and improving data accuracy by reducing manual errors by 30%.

Key Projects

Skin Cancer Detection Using Deep Learning and GANs

- Co-authored a peer-reviewed Springer book chapter on scalable cloud-based medical imaging pipelines, increasing research accessibility for 500+ clinicians.
- Augmented dataset size by 150% using SRGAN and DCGAN to synthesize dermoscopic images, enhancing model generalizability.
- Boosted classification accuracy by 10% via data augmentation and hyperparameter tuning of CNNs using TensorFlow.
- Evaluated model robustness with 10-fold cross-validation and ensemble learning, ensuring consistent performance across varied image conditions.

Face Mask Detection Using Computer Vision

- Invented and patented a real-time “System and Method for Detecting a Face Mask”, improving public health surveillance and safety compliance.
- Trained and optimized a MobileNetV2 model on custom datasets, achieving 90%+ accuracy using TensorFlow and OpenCV.
- Reduced model latency by 25% via quantization techniques and deployment on resource-efficient edge devices.
- Deployed real-time alert systems and dashboards, improving campus-wide mask compliance by 30%.

Fire Detection System Using Deep Learning

- Developed a CNN-based fire detection solution with AWS Lambda integration, enabling rapid response through edge deployment.
- Reached 90% detection accuracy and minimized false positives by 20% using refined image preprocessing and CNN tuning.
- Automated fire alerts via IoT temperature sensors, triggering warnings within 5 seconds of detection.
- Simulated fire scenarios to test model reliability under occlusions and noisy conditions, validating system robustness.

Technical Skills

Programming: Python, R, SQL (PostgreSQL, Oracle SQL), C++

Frameworks/Libraries: TensorFlow, Keras, Scikit-learn, PySpark, OpenCV

Tools & Platforms: AWS, Apache Airflow, Kafka, Snowflake, Git, Docker

Visualization: Tableau, Power BI, QlikView, SAS Visual Analytics

Data Engineering: ETL Pipelines, Data Warehousing, API Integration, Batch/Stream Processing